HDOC DAY 60

PROBLEM

Dear Student,

The **HDOC Day-60** activity for Batches 15 & 16 is attached herewith.

Write a C program that reads two strings, s and t. Assume that both strings contain only lowercase characters. Pass the strings to a function named check_anagram() using their addresses. Use pointers to pass the strings between the calling function and check_anagram(). The function check_anagram() must:

- Determine whether s and t contain the same characters with the same frequencies.
- Print "Anagram" if the two strings are valid anagrams.
- Print "Not Anagram" if the two strings are not valid anagrams.
- Treat the order of characters as irrelevant.
- Consider strings of different lengths as not anagrams.
- Use a character-frequency-counting approach to solve the problem.

CODE

```
#include <stdio.h>

#include <string.h>


void check_anagram(char *s, char *t)

{

   int freq[26] = {0};


   if (strlen(s) != strlen(t)) {

      printf("Not Anagram\n");

      return;
```

```
    }

    while (*s) {
        freq[*s - 'a']++;
        s++;
    }

    while (*t) {
        freq[*t - 'a']--;
        t++;
    }

    for (int i = 0; i < 26; i++) {
        if (freq[i] != 0) {
            printf("Not Anagram\n");
            return;
        }
    }

    printf("Anagram\n");
}

int main(void)
{
    char s[100], t[100];

    printf("Enter first string: ");
    scanf("%99s", s);
```

```
    printf("Enter second string: ");
    scanf("%99s", t);

    check_anagram(s, t);

    return 0;
}
```